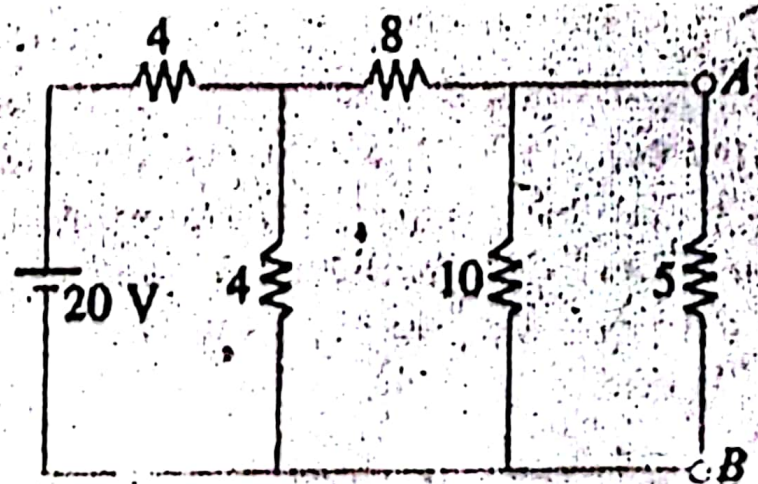


Chapter 2: Network theorem

1. State and explain Kirchhoff's current law.[2019]
Or, State and explain KCL[2021]
2. State and explain Kirchhoff's voltage law.
Or, State and explain KVL[2021]
3. Thevenin's theorem.
4. Norton's theorem.[2019]
5. Maximum power transfer theorem.
6. How to Thevenize a Given Circuit ? (2010)
7. Describe Power Transfer Efficiency
8. State and explain Superposition Theorem [2022]
9. Describe Delta/Star* Transformation
10. State Norton's theorem . Calculate the current through 5- Ω resistor of the following circuit using Norton's theorem. [2021]



Chapter 3: Introduction to Semiconductors

1. What is semiconductor material?[2018] Classify semiconductor materials.[2010]
2. What is semiconductor? Explain the effect of temperature on semiconductor.[2019]
3. describe the classification of semiconductors.[2019,2021 Oper]
4. Explain how n-type and p-type semiconductors are produced. [2010,2011,2018]
Or, Describe how n-type and p-type materials are formed [2021]
- Or, Explain the doping Process of extrinsic semiconductor.[2022]
5. Explain how depletion layer is developed in a PN junction. [2010]
Or, Describe what happen when a P-type and N-type material are brought together hence define barrier voltage.[2015]
6. Describe the operation of a PN junction with characteristic curve.
[2010,2013,2014,2018]
7. What is barrier voltage of a diode? Explain how the barrier voltage is developed in it.
[2011,2013]
8. What do you understand by intrinsic and extrinsic semi-conductor?
[2011,2013,2014,2016,2020]
9. What is doping?[2011,2016,2022]
10. Distinguish between P-type and N-type material.[2015]
11. Define Semi-conductor, Rectifier, Clipper.[2015,2020]
12. Describe the operation of a P-N junction diode when it is:-
[2013,2014,2015,2016,2020,2022]
 No bias;
 Forward bias;
 Reverse bias with I-V curve.
13. Define the terms conductor, semiconductor, insulator in terms of energy band diagram. [2016,2018,2021]
Or, Explain the classification of materials according to energy band.[2022]
14. Explain the n-type semiconductor with energy band diagram.[2019]
15. What is biasing? Why biasing is necessary for a transistor circuit.[2019]
16. Describe the operation of the P-N junction diode with V-I characteristics.[2019]
17. Explain how p-type semiconductor are produced with energy band diagram. (2020)
18. What is P-N Junction? Describe I-V characteristics of a forward biased P-N Junction.
(2021)
19. Define Clipper and Clamper circuit (2021)